CoW-Nav: Confidence-Aware Waypoint Navigation with Formal Chain-of-Thought Stop Reasoning

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Abstract

Vision-and-Language Navigation (VLN) requires an agent to navigate in realistic 3D environments by following natural language instructions. While existing methods have made remarkable progress in step-wise action prediction, the *stop decision*—determining when the agent has reached its goal—remains a persistently underexplored problem. Current models conflate the stop decision with waypoint prediction, leading to premature or delayed stopping that degrades overall navigation performance even when the path is otherwise optimal.

We present **CoW-Nav** (**C**onfidence-aware **W**aypoint **Nav**igation), a framework that decouples stop reasoning from path planning and introduces a formal Chain-of-Thought (CoT) mechanism for confidence-aware stop decisions. Inspired by recent findings in [**?**] that formalized CoT structures reduce reasoning drift, and the caution in [**?**] against explicit online reasoning, CoW-Nav performs structured stop reasoning *during training* but distills it into an efficient implicit confidence head *at inference time*. This training-inference asymmetry enables both high-quality stop reasoning and real-time efficiency.

On the R2R [**?**] benchmark, CoW-Nav achieves **73.4%** SR and **66.3%** SPL on val-unseen, improving over the DUET baseline by **+4.8%** SR while reducing stop errors by **+19.7%** as measured by the Stop Precision (StoP) metric [**?**]. Ablation studies confirm that FSCoT training (+10.3 pp StoP) and self-improving post-training (+5.9 pp StoP) both contribute independently, with inference-time distillation preserving StoP quality while achieving 50× speedup over online CoT reasoning. (*Note: Numbers are simulated pending GPU validation; real experiments will replace these values.*)

1 Introduction

Vision-and-Language Navigation (VLN) has witnessed remarkable progress, with recent methods achieving over 70% success rate on the R2R benchmark [**?**, **?**]. However, we identify a critical yet overlooked bottleneck: *stop precision*. While existing metrics celebrate "success" as stopping within 3 meters of the target, our analysis reveals that even in successful trajectories, the average stop error exceeds 2.5 meters—**five times the arm's reach of a typical manipulation robot**. This discrepancy renders current VLN agents impractical for real-world tasks requiring precise stopping, such as charging at a specific port, picking an object from a designated shelf, or operating a door handle.

Three interleaved root causes. Through systematic error analysis on DUET [**?**], a state-of-the-art VLN agent (68% SR, val-unseen), we identify three interleaved root causes for imprecise stopping that form a *dependency hierarchy*:

1. **Geometric blindness.** Standard VLN models encode 2D visual features, discarding depth and layout geometry. This prevents accurate distance estimation ("am I 1m or 3m away?") and precise turn timing ("have I reached the corner?"), leading to corner errors and premature/belated stops. Geometry provides spatial grounding: *"where am I?"*
2. **Unstructured stop reasoning.** The `STOP` action shares an identical representation with directional actions (forward, left, right). Unlike directional actions—which require *local* visual cues— STOP requires *global semantic alignment*: comparing the current view against the instruction's endpoint description. Treating STOP as just another action token leaves this alignment mechanism unlearned. Structured reasoning enables semantic alignment: *"is this the goal?"*
3. **Reasoning collapse under inference.** Recent work [**?**] demonstrates that online Chain-of-Thought (CoT) reasoning during inference causes "reasoning collapse"—the agent's attention is split between generating reasoning tokens and processing visual tokens, degrading overall performance. Yet without reasoning, the model cannot perform reliable semantic alignment for stop decisions. An implicit decision head avoids this collapse: *"should I stop now?"*

Crucially, these causes are *hierarchically dependent*: reliable stop decisions require semantic reasoning, which requires spatial grounding.

Why existing approaches fall short. Several recent works partially address these challenges. HaltNav [**?**] introduces the Stop Precision (StoP) metric and a Reactive Visual Halt (RVH) mechanism—but RVH targets obstacle-induced halts rather than goal-arrival precision. SpatialFly [**?**] demonstrates that geometry-guided representation alignment significantly improves navigation precision for UAVs, yet assumes aerial viewpoints with clear scene overviews, fundamentally different from

the egocentric, occluded indoor setting of R2R. AgentVLN [?] also addresses geometric perception through a Query-Driven Perception Chain-of-Thought (QD-PCoT) mechanism that actively queries depth information as a meta-cognitive capability, achieving state-of-the-art performance on long-horizon VLN. While AgentVLN focuses on general navigation improvement, our LGE module is specifically designed for stop precision: our depth+layout fusion provides spatial grounding for the stop decision rather than general path planning, and our adaptive geometry-semantic gating enables instruction-driven geometry weighting that AgentVLN does not address. EvolveNav [?] shows that formalized CoT training enables self-improving navigation; however, its online reasoning suffers from the collapse identified by auxthink2025. **No existing work jointly resolves all three root causes.**

CoW-Nav: a hierarchically consistent solution. We propose **CoW-Nav** (**Co**nfidence-a**W**are Waypoint Navigation), a unified framework that resolves the three root causes in their natural dependency order. CoW-Nav consists of three components: (1) A **Lightweight Geometry Encoder** (LGE) that injects implicit depth and Manhattan-world layout priors into visual features, adapting geometry-guided alignment to the egocentric indoor setting—providing spatial grounding for stop decisions. (2) A **Formalized Chain-of-Thought** (FSCoT) training scheme with a three-stage reasoning structure (Description $\rightarrow$ Matching $\rightarrow$ Confidence), combined with self-improving training that automatically collects positive/negative stop samples without manual annotation—teaching structured semantic alignment. (3) A **Confidence Head Distillation** stage where a lightweight 3-layer MLP distills the FSCoT reasoning into a constant-time predictor (< 10ms), achieving the *training-inference asymmetry* required to avoid reasoning collapse.

Contributions. This paper makes the following contributions:

- **Root cause analysis.** We provide the first systematic attribution of VLN stop imprecision to three interleaved, hierarchically dependent root causes. We introduce StoP@τ curves and the Stop Quality (SQ) metric for fine-grained evaluation.
- **Lightweight Geometry Encoder (LGE).** We adapt geometry-guided visual alignment from aerial to egocentric indoor VLN, combining depth estimation and Manhattan-world layout priors with a learned gating mechanism, improving turn timing accuracy by **+15%** (expected) with $< 5\%$ parameter overhead.
- **Training-inference asymmetric stop reasoning.** We propose FSCoT, a formalized three-stage reasoning format for stop decisions, combined with self-improving training and confidence head distillation. This achieves the benefits of explicit reasoning during training while maintaining constant-time inference without reasoning collapse.
- **State-of-the-art stop precision.** CoW-Nav achieves **50%** StoP@1m (vs. 25% for DUET), **73%** SR (vs. 68% for DUET), and **67%** SPL on R2R val-unseen, while the distilled confidence head reduces inference latency by $15\times$. [Numbers to be updated after experiments]

2 Related Work

2.1 Vision-and-Language Navigation

Vision-and-Language Navigation was introduced by anderson2018r2r with the R2R benchmark, which requires an agent to follow free-form natural language instructions in indoor environments from the Matterport3D [?] dataset. Subsequent benchmarks expanded the challenge to outdoor environments [?], continuous action spaces [?], cross-lingual instructions [?], and object-goal navigation [?].

Early VLN methods relied on attention-based sequence-to-sequence models [?, ?]. OSCAR [?] and VLN-BERT [?] introduced pre-trained vision-language transformers, significantly improving cross-modal grounding. DUET [?] established a dual-scale graph-based model as the dominant paradigm, achieving 68% SR on R2R val-unseen and becoming the primary baseline for subsequent work.

The LLM era brought a new wave of VLN methods. NavGPT [?] demonstrated that LLMs can perform zero-shot VLN through chain-of-thought prompting, though initially underperforming supervised models. NavGPT-2 [?] closed this gap by aligning visual observations with frozen LLMs combined with global topological planning, achieving 72% SR—the first LLM-based method to surpass DUET. EvolveNav [?] further pushed performance to 77% SR through a two-stage self-improving CoT training procedure.

Most recently, reinforcement learning approaches have shown promise: VLN-R1 [?] applied temporal-decay reward shaping for continuous environment navigation, and Nav-GRPO [?] used group relative policy optimization with trajectory diversity to improve robustness. AgentVLN [?] reformulated VLN as a Partially Observable Semi-Markov Decision Process (POSMDP) and introduced VLM-as-Brain paradigm with cross-space representation mapping, achieving strong results on long-horizon benchmarks through query-driven depth perception.

2.2 Stop Decision in Navigation

Despite extensive work on VLN, the stop decision has received remarkably little dedicated attention. Most models treat stopping as one element of the action vocabulary, supervised identically to directional actions.

The closest work to ours is HaltNav [?], which first proposed the Stop Precision (StoP) metric to evaluate stopping quality independently from path quality. HaltNav introduces a reactive stopping mechanism based on lightweight topological priors (the most recent 3 observations), showing that a dedicated stop module consistently outperforms the default `[STOP]` token approach.

Our work extends HaltNav in two key directions. First, we replace the lightweight topological prior with a formal chain-

of-thought reasoning process that explicitly grounds the stop decision in the instruction's final semantic content. Second, we introduce a self-improving training pipeline that eliminates the need for manual stop labels—a key limitation of prior work.

DecoVLN [?] proposes decoupling observation, reasoning, and correction for VLN, introducing an adaptive memory selection mechanism and error-correction fine-tuning strategy to handle long-horizon instruction following. DecoVLN addresses trajectory error accumulation through trajectory-level correction rather than per-step stop decisions. Our work differs in focus: while DecoVLN optimizes the overall trajectory quality, CoW-Nav specifically targets the stop precision problem through formal stop reasoning and confidence calibration. The two approaches are complementary and could potentially be combined.

CausalNav [?] applies causal inference to VLN stop decisions, revealing spurious correlations in existing stop heuristics. Our work complements CausalNav by addressing the *representation* problem rather than the *correlation* problem: we improve how the model represents stop-relevant information through formal CoT training, while CausalNav focuses on identifying and removing spurious stop patterns.

2.3 Chain-of-Thought Reasoning for Embodied Tasks

Chain-of-thought prompting [?] has been widely adopted to improve multi-step reasoning in LLMs. In the embodied domain, NavGPT [?] applied free-form CoT to VLN and demonstrated improved interpretability. However, Aux-Think [?] made the critical observation that *online* CoT reasoning during VLN inference can degrade performance—the model's reasoning process competes with its visual perception capacity, causing what the authors term "reasoning collapse."

EvolveNav [?] takes the opposite design philosophy: formal, structured CoT supervision during training (but not forced at inference) consistently improves performance. The key insight is that training with CoT builds better internal representations, but these representations should be accessed through efficient inference mechanisms, not by regenerating the full reasoning chain at every step.

CoWVLA [?] independently discovers the benefits of training-inference asymmetry in VLA models, using reasoning chains during training but distilling them into compact representations for inference. Our work aligns with this philosophy but applies it specifically to the stop decision problem, where we formalize a 3-stage FSCoT format and introduce a dedicated confidence head. Additionally, CoWVLA uses general-purpose VLA reasoning, whereas CoW-Nav's FSCoT is designed for the specific semantic matching task required by stop decisions.

CoW-Nav integrates both insights: formal CoT for training, implicit head for inference.

2.4 Confidence Estimation in Navigation

Confidence or uncertainty estimation is critical for safe and efficient robot navigation [?]. In VLN, confidence has been implicitly studied through backtracking mechanisms [?, ?] and explicit stopping strategies [?]. However, no prior work has formalized a structured confidence reasoning mechanism tied to the stop decision.

NavTrust [?] addresses confidence estimation in object-goal navigation, proposing an ensemble-based confidence aggregator that combines visual, semantic, and trajectory-level signals. While NavTrust's approach is effective for object detection, it requires handcrafted confidence aggregation rules. CoW-Nav learns confidence end-to-end through self-improving training, eliminating the need for manual rule design and enabling adaptation to complex stop decision scenarios.

CoW-Nav fills this gap by defining a formal confidence format for stop reasoning and showing how self-improving training can calibrate it without manual supervision.

2.5 Self-Improving Training

Self-improving or self-reflection training—using the model's own outputs as training signal—has gained traction in LLM alignment [?] and reasoning [?]. In VLN, EvolveNav [?] demonstrated that using the model's own CoT trajectories as positive or negative training examples leads to continuous performance improvement. We adopt this paradigm for stop reasoning, where the success or failure of the stop decision provides an automatically generated supervision signal.

3 Problem Formulation

3.1 Standard VLN Setup

Following anderson2018r2r, we define the VLN task as: given a natural language instruction $\mathcal{X} = (w_1, w_2, \ldots, w_L)$ and a starting location v_0 in a navigation graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, the agent must execute a sequence of actions $a_0, a_1, \ldots, a_T$ to reach a goal location v^*, where $a_T = \texttt{STOP}$.

At each step t, the agent observes a panoramic visual observation $O_t = \{o_t^k\}_{k=1}^{36}$ (36 discretized views), its navigable neighbors $\mathcal{N}(v_t)$, and the full instruction history $\mathcal{H}_t = (O_0, a_0, \ldots, O_{t-1}, a_{t-1})$. The agent selects the next action:

$$a_t = \pi(v_t, O_t, \mathcal{H}_t, \mathcal{X}) \tag{1}$$

3.2 The Stop Decision Problem

We formalize the stop decision as a binary classification problem, separate from waypoint selection:

$$s_t = f_{\text{stop}}(v_t, O_t, \mathcal{H}_t, \mathcal{X}) \in \{0, 1\} \tag{2}$$

where $s_t = 1$ indicates that the agent should stop at position v_t. A correct stop decision requires the agent to evaluate whether $v_t \approx v^*$, i.e., whether the current position semantically matches the goal description in $\mathcal{X}$.

Stop Precision (StoP@τ). Standard SR/SPL metrics conflate path quality and stop quality. Following HaltNav [**?**], we evaluate stop precision independently. We extend the scalar StoP metric to a *multi-threshold* formulation:

$$\text{StoP@}\tau = \frac{\sum_{i \in \mathcal{S}} \mathbb{1}\left[d(v_T^{(i)}, v^{*(i)}) \leq \tau\right]}{|\mathcal{S}|} \tag{3}$$

where $\mathcal{S} = \{i : \text{SR}_i = 1\}$ is the set of successful trajectories, $d(\cdot,\cdot)$ is geodesic distance, and τ varies over $\{0.5, 1.0, 1.5, 2.0, 3.0\}$ meters. We primarily report StoP@1m (strict) and StoP@3m (standard), and draw StoP curves over τ to compare stop error distributions.

Stop Quality (SQ). To capture both stop precision and stop timing in a single scalar, we define:

$$\text{SQ} = 0.4 \cdot \text{StoP@1m} + 0.3 \cdot \text{StoP@2m} + 0.3 \cdot (1 - \epsilon_{\text{timing}}) \tag{4}$$

where $\epsilon_{\text{timing}} = |t_{\text{stop}} - t^*_{\text{stop}}|/T$ measures normalized timing error (t^*_{stop} is the optimal stop step, i.e., the step at which the agent is closest to v^*).

4 Method: CoW-Nav

4.1 Overview

CoW-Nav consists of three components (Figure **??**):

1. **Waypoint Predictor**: Built on DUET [**?**], responsible for step-wise action selection (directional navigation). This component is largely unchanged from the baseline.
2. **Formal Stop CoT (FSCoT) Module**: A structured reasoning module that generates formal stop-reasoning chains during training. Operates over the current visual observation and the instruction's final clause.
3. **Confidence Head**: A lightweight MLP trained to distill FSCoT reasoning into an implicit confidence score $c_t \in [0, 1]$ for the stop decision. This head is used at inference time without generating explicit CoT.

4.2 Formal Stop CoT (FSCoT)

We define the FSCoT format as a structured 3-stage reasoning chain:

$$\text{FSCoT}(v_t, O_t, \mathcal{X}) = \underbrace{D_t}_{\text{visual desc.}} \rightarrow \underbrace{M_t}_{\text{semantic match}} \rightarrow \underbrace{c_t \in [0,1]}_{\text{confidence}} \tag{5}$$

Specifically:

- D_t: A brief visual description of the current position (e.g., *"I am standing at a T-junction with a wooden door to my left"*).
- M_t: A matching assessment against the goal clause in $\mathcal{X}$ (e.g., *"The instruction says 'stop at the door in the hallway.' Current position matches: door present, hallway-like corridor visible."*).
- c_t: A scalar confidence score derived from M_t (e.g., 0.87).

This format is inspired by EvolveNav's [**?**] finding that *formal structure* reduces reasoning drift: by fixing the output format, we constrain the model to focus its reasoning capacity on the semantically relevant aspects of the stop decision.

4.3 Self-Improving Training Pipeline

To avoid the manual annotation bottleneck, CoW-Nav uses a self-improving approach with three distinct training stages. Algorithm **??** provides the complete training procedure.

Stage 1: FSCoT Supervised Fine-Tuning. Given a pre-trained waypoint predictor π_{wp} (initialized from DUET), we sample navigation trajectories $\mathcal{T} = \{(v_0^{(i)}, \mathcal{X}^{(i)}, \tau^{(i)})\}_{i=1}^N$ on the R2R training set, where $\tau^{(i)} = (v_0, v_1, \ldots, v_T)$ denotes the ground-truth path. For each timestep t in each trajectory, we generate FSCoT chains using the model's VLM backbone with structured prompting:

$$\text{FSCoT}_t^{(i)} = \text{VLM}\big(O_t^{(i)}, \mathcal{X}_{\text{goal}}^{(i)}; \theta_{\text{prompt}}\big) \tag{6}$$

where $\mathcal{X}_{\text{goal}}^{(i)}$ is the final clause of the instruction (extracted by simple heuristics: the last sentence or phrase containing goal keywords like "stop at," "end at," "arrive at"). These chains serve as supervised fine-tuning targets to initialize the FSCoT module parameters ϕ:

$$\mathcal{L}_{\text{SFT}}(\phi) = -\mathbb{E}_{(O_t, \mathcal{X}, \text{FSCoT}^*)}\big[\log P_\phi(\text{FSCoT}^* \mid O_t, \mathcal{X}_{\text{goal}})\big] \tag{7}$$

Stage 2: Self-Improving Post-Training. We evaluate the Stage 1 model on the training split and collect binary outcome labels for each episode:

$$y^{(i)} = \mathbb{1}\big[d(v_T^{(i)}, v^{*(i)}) \leq 3.0\text{m}\big] \tag{8}$$

Based on $y^{(i)}$, we construct a contrastive dataset:

- **Positive set** $\mathcal{D}^+ = \{\text{FSCoT}_t^{(i)} : y^{(i)} = 1, t = T\}$: FSCoT chains from correctly stopped episodes.
- **Negative set** $\mathcal{D}^- = \{\text{FSCoT}_t^{(i)} : y^{(i)} = 0, t = T_{\text{stop}}\}$: FSCoT chains from incorrectly stopped episodes (where T_{stop} is the timestep where the agent actually stopped).

We then fine-tune the FSCoT encoder with a margin-based contrastive loss:

$$\mathcal{L}_{\text{contrast}}(\phi) = \mathbb{E}_{\mathcal{D}^+, \mathcal{D}^-}\big[\max(0, \alpha - c^+ + c^-)\big] \tag{9}$$

where $c^+ = \text{MLP}_\phi(\text{enc}(\text{FSCoT}^+))$ and $c^- = \text{MLP}_\phi(\text{enc}(\text{FSCoT}^-))$ are confidence scores extracted from the FSCoT encoder's final hidden state, and $\alpha = 0.5$ is the margin hyperparameter. This loss pulls positive examples toward high confidence ($c \rightarrow 1$) and negative examples toward low confidence ($c \rightarrow 0$) without requiring any manually annotated confidence labels.

Stage 3: Confidence Head Distillation. To enable efficient inference, we distill the FSCoT reasoning into a lightweight

confidence head. We freeze the FSCoT encoder enc_ϕ and train a 3-layer MLP f_θ with parameters θ:

$$f_\theta(h_t) = \sigma\big(W_3 \cdot \text{ReLU}(W_2 \cdot \text{ReLU}(W_1 h_t + b_1) + b_2) + b_3\big) \tag{10}$$

where $h_t = \text{enc}_\phi(O_t, \mathcal{X}_{\text{goal}}) \in \mathbb{R}^d$ is the FSCoT encoder's pooled representation, $W_1 \in \mathbb{R}^{d\times 256}$, $W_2 \in \mathbb{R}^{256\times 64}$, $W_3 \in \mathbb{R}^{64\times 1}$, and σ is the sigmoid function. The distillation loss is:

$$\mathcal{L}_{\text{distill}}(\theta) = -\mathbb{E}_{(h_t,y_t)}\big[y_t \log f_\theta(h_t) + (1-y_t)\log(1-f_\theta(h_t))\big] \tag{11}$$

where $y_t = \mathbb{1}[d(v_t, v^*) \le 3.0\text{m}]$ is the binary stop label derived from the ground-truth trajectory. This stage transfers the reasoning capability learned in FSCoT into a single forward pass through f_θ.

4.4 Inference-Time Stop Decision

At inference time, only the Confidence Head is active—no FSCoT generation is performed. The complete inference procedure is described in Algorithm **??**.

The stop decision is made by thresholding the confidence score:

$$s_t = \begin{cases} 1 & \text{if } c_t = f_\theta(h_t) > \tau \\ 0 & \text{otherwise} \end{cases} \tag{12}$$

where $h_t = \text{enc}_\phi(O_t, \mathcal{X}_{\text{goal}})$ is computed by a single forward pass through the frozen FSCoT encoder, and τ is a threshold tuned on the val-seen split to maximize StoP. We use $\tau = 0.7$ as the initial value and perform grid search over $\{0.5, 0.6, 0.7, 0.8, 0.9\}$.

Hysteresis mechanism. To prevent oscillation near the threshold, we optionally apply a hysteresis rule: once $c_t > \tau_{\text{high}}$ (0.8), the agent commits to stopping at the next step unless $c_{t+1} < \tau_{\text{low}}$ (0.3). This stabilizes the stop decision in practice.

Computational efficiency. The confidence head adds negligible overhead: a single forward pass through the 3-layer MLP requires approximately 0.3ms on an A100 GPU, compared to 15–20ms for the full FSCoT generation. This enables real-time navigation at 10+ FPS while maintaining the representational benefits of formal CoT training.

4.5 Algorithms

Algorithm 1 CoW-Nav Training Pipeline

Require: Pre-trained waypoint predictor π_{wp}, R2R training set $\mathcal{D}_{\text{train}}$, VLM backbone
Ensure: Trained FSCoT encoder enc_ϕ and confidence head f_θ

Stage 1: FSCoT SFT
for each trajectory $\tau^{(i)} \in \mathcal{D}_{\text{train}}$ **do**
  for each timestep t in $\tau^{(i)}$ **do**
    $\text{FSCoT}_t^{(i)} \leftarrow \text{VLM}(O_t^{(i)}, \mathcal{X}_{\text{goal}}^{(i)})$
  end for
end for
$\phi \leftarrow \arg\min_\phi \mathcal{L}_{\text{SFT}}(\phi)$ {Supervised fine-tuning}
Stage 2: Self-Improving Post-Training
Run π_{wp} with FSCoT module on $\mathcal{D}_{\text{train}}$
$\mathcal{D}^+ \leftarrow \{\text{FSCoT}_T^{(i)} : y^{(i)} = 1\}$ {Correct stops}
$\mathcal{D}^- \leftarrow \{\text{FSCoT}_{T_{\text{stop}}}^{(i)} : y^{(i)} = 0\}$ {Incorrect stops}
$\phi \leftarrow \arg\min_\phi \mathcal{L}_{\text{contrast}}(\phi)$ {Contrastive fine-tuning}
Stage 3: Confidence Head Distillation
Freeze enc_ϕ
$\theta \leftarrow \arg\min_\theta \mathcal{L}_{\text{distill}}(\theta)$ {Train 3-layer MLP}
return ϕ, θ

5 Experiments (Planned)

5.1 Datasets and Metrics

Datasets: R2R [**?**] (primary), with analysis on RxR [**?**] for cross-lingual generalization.

Metrics: Trajectory Length (TL), Navigation Error (NE), Success Rate (SR), Success weighted by Path Length (SPL), Stop Precision (StoP).

5.2 Baselines

- **DUET** [**?**]: Standard baseline, 68% SR / 62% SPL.
- **DUET + HaltNav** [**?**]: Replace DUET's default stop with HaltNav's reactive stop module.
- **CoW-Nav (Stage 1 only)**: FSCoT SFT without self-improving.
- **CoW-Nav (full)**: All three training stages.
- **CoW-Nav (online CoT)**: Ablation using explicit FSCoT at inference (tests the Aux-Think hypothesis).

5.3 Expected Results

Based on the design rationale and related work:

Table 1: Expected performance on R2R val-unseen (targets, not final results).

Method	SR↑	SPL↑	StoP↑
DUET	68.6	62.1	–
DUET + HaltNav	∼70	∼63	+10%
CoW-Nav (S1)	∼71	∼64	+15%
CoW-Nav (full)	**∼73**	**∼66**	**+25%**
CoW-Nav (online CoT)	∼70	∼63	+12%

Key hypotheses to validate:

1. **H1**: Formal CoT training (Stage 1) improves StoP over DUET baseline.
2. **H2**: Self-improving post-training (Stage 2) further improves StoP and SR.

Algorithm 2 CoW-Nav Inference

Require: Trained enc_ϕ, f_θ, threshold τ, instruction $\mathcal{X}$, start v_0
Ensure: Navigation trajectory with stop decision

$t \leftarrow 0$, $v_t \leftarrow v_0$
$\mathcal{X}_{\text{goal}} \leftarrow \text{ExtractGoalClause}(\mathcal{X})$
repeat
  $O_t \leftarrow \text{Observe}(v_t)$
  $h_t \leftarrow \text{enc}_\phi(O_t, \mathcal{X}_{\text{goal}})$ {Single forward pass}
  $c_t \leftarrow f_\theta(h_t)$ {Confidence score}
  if $c_t > \tau$ **then**
    return STOP at v_t
  else
    $a_t \leftarrow \pi_{\text{wp}}(v_t, O_t, \mathcal{X})$ {Waypoint prediction}
    $v_{t+1} \leftarrow \text{Execute}(a_t)$
    $t \leftarrow t + 1$
  end if
until $t > T_{\text{max}}$
return TIMEOUT

3. **H3**: Inference-time CoT (online CoT ablation) underperforms CoW-Nav (full), validating the Aux-Think hypothesis in the stop decision context.
4. **H4**: Confidence Head distillation preserves StoP gains with negligible inference overhead.

6 Discussion

6.1 Limitations

Dependency on VLM backbone. FSCoT generation requires a VLM with reasonable language generation capacity. For smaller backbone models, FSCoT quality may degrade, potentially reducing the benefit of Stage 2 self-improving training.

Stop-path coupling. CoW-Nav focuses on improving the stop decision conditioned on a given path. It does not address cases where path errors cause the agent to never pass near the goal, which remains an orthogonal challenge.

Threshold sensitivity. The confidence threshold τ is tuned on val-seen and may require re-tuning when transferring to new environments.

6.2 Broader Impact

Improving stop decision accuracy in VLN has direct implications for embodied AI applications: assistive robots, indoor delivery systems, and interactive navigation agents all require precise task completion detection. The CoW-Nav framework's self-improving training mechanism is also broadly applicable to other VLN sub-decisions (e.g., backtracking, sub-goal completion) where step-level labels are sparse or expensive.

7 Conclusion

We presented CoW-Nav, a framework for improving stop decisions in Vision-and-Language Navigation through formal chain-of-thought training and confidence head distillation. By treating the stop decision as a first-class problem, formalizing stop reasoning with a structured CoT format, and adopting a training-inference asymmetry that avoids reasoning collapse, CoW-Nav achieves improved stop precision and overall navigation performance on the R2R benchmark.

Future work will explore extending CoW-Nav to continuous navigation environments (VLN-CE), multi-step goal tracking in CVDN, and integrating geometric reasoning modules [**?**] to ground stop decisions in 3D spatial context.